Fire-Aware Robot Dogs

Machine-Learning Wildfire Forecasts for Safe Navigation and Scouting

Shawn Zeng · Monta Vista High School

ONGOING WORK · CURRENTLY IN SIMULATION

built & tested in simulation
proposed robot-dog use

1 Predict

The model forecasts **when** each 50 m cell will catch fire. Designed to run on the dog or its controller.

2 Safe route

Plan paths that avoid cells predicted to ignite **before the dog gets through**.

3 Escape path

Always keep a way out that the forecast says **will stay clear**.

4 Scout

Check the fire edge and the **planned containment line** before crews commit.

5 Feedback

Send back real fire position, wind and fuel so **the next forecast is better**.



robot dog

Know where the fire *will* be – not just where it is.

- **Safer crews** — Fewer firefighters sent into danger to look.
- **Robot stays safe** — Routes and exits keep it from getting trapped.
- **Smarter sensing** — Measure where it changes decisions.